An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

Finding people in floodwater faster than a boat can

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1. Executive summary

Floods displace millions of people in India every year. The slow part of a rescue is usually not reaching survivors — it is *finding* them.

We are building three one-metre, 6.4 kg multirotor aircraft that fly a search pattern over flooded ground as a coordinated group under **a single operator**, recognise people from the air, report their coordinates to within about a metre, and drop a small relief kit to each. It works with **no mobile network and no internet** — which matters, because those are among the first things a flood takes away.

The design is further along than it may look. The sizing, the error budgets and the flight software are written down, reproducible, and honest about which numbers are measured and which are still calculated. Coordination and failsafe behaviour have been demonstrated across three simultaneous simulated autopilots.

Two things remain. **No aircraft has flown yet**, and while checking our own arithmetic we found a real flaw in the vision pipeline: the camera sees a survivor clearly and our software throws that detail away before the detector looks at it. We have a fix, and we can test it on free public data before committing to anything.

This document sets out the problem, the system, what it delivers, who benefits and what it costs — **INR 7,34,379 across nine small steps**, the first of which is INR 31,371.

2. The problem we are solving

After a flood, the binding constraint is time-to-locate.

Ground and boat teams reach people, but they search slowly, at high risk to themselves, and they cannot see over debris fields or onto rooftops. Helicopters search quickly but are scarce, expensive per flight hour, and usually committed to evacuation rather than systematic survey. In between sits a gap that nothing currently fills: *covering a large area quickly and reporting exactly where people are*.

Small drones look like the obvious answer and mostly do not fill it either, for four reasons:

- **One pilot per aircraft.** Most systems need a trained operator flying manually. Three aircraft means three pilots, and a disaster response rarely has three to spare.
- **They rely on a network.** Cloud detection and mobile telemetry are useless in exactly the conditions that produced the emergency.

- **They find, but do not help.** Spotting someone and reaching them are different problems, and a drone that only spots leaves a gap of hours.
- **Seeing a person from the air is genuinely hard.** This is the part most proposals skip, and it is where we found our own mistake.

The technical problem inside that

A person in floodwater shows only head and shoulders — roughly 40 cm across. Our camera at 60 m resolves that across about 26 pixels. But detection models take a fixed, small input, so the picture is shrunk to fit, and after shrinking the person occupies about **13 pixels**.

At that size, published results for detecting tiny people report accuracy between 29 % and 61 %. That is not good enough when a missed person is a missed rescue. And the obvious response — use a smaller, faster model — makes it *worse*, because smaller models take smaller inputs.

Three further things make floods harder than they first appear:

- **No suitable training data exists.** Flood datasets label buildings and water damage, not people. Datasets of people in water are open sea: clear blue water, no debris, no rooftops.
- **Thermal cameras do not rescue us.** In water, skin cools toward water temperature and the heat difference largely disappears. Thermal helps on rooftops and fails in exactly the case we care most about.
- **We were sizing for the wrong target.** Our early work assumed a person lying flat, 1.7 m long, rather than a person in water, 0.4 m across. That made the problem look four times easier than it is. We found the mistake ourselves — and it is precisely why we now want to measure rather than assume.

3. Our solution

The system

Three identical aircraft, one operator, no network dependency.

- **The area is divided automatically.** The operator draws a search region of any shape; the software splits it into three balanced pieces and plans a lawnmower path over each. We have verified this on thirteen awkward boundary shapes, including ones with notches and slots, with no path leaving the region and under 0.02 % workload imbalance.
- **The aircraft fly themselves.** Once launched they follow the plan without further instruction. Our test harness proves this structurally, by showing that no command is transmitted after the mission starts.
- **Detection happens on board.** Each aircraft carries a small computer and an AI accelerator, so nothing depends on a radio link or a server.
- **Positions are reported to about a metre**, using centimetre-accurate satellite positioning corrected from a base station we set up ourselves.
- **Each aircraft carries four relief kits** and can descend, stop over a survivor, and release one.
- **It fails safely.** Loss of link, low battery or an operator abort all converge on the same return-and-land behaviour, verified in simulation.

The fix for the vision problem

Three changes, in order of how much they help.

1. Fly lower. Coming down from 60 m to 30 m more than triples how large a person appears. It costs a little flying time, and we have a great deal to spare — the planned mission takes under eight minutes.

2. Stop shrinking the picture. Instead, cut the full-resolution image into overlapping tiles and examine each one properly. This is well-established practice; the catch is that it costs four times the computing power.

3. Make step 2 affordable with a cheap first look. Most of a flooded area is empty water. A small, fast model checks each tile and asks only *"could there be anything here?"* The expensive detector then runs only where the answer is yes. About **87 % of tiles are empty** and we need to skip **80 %** to break even, so the arithmetic closes.

Steps 1 and 2 together take a person from **13 pixels to 52** — **sixteen times the area** — for the same computing cost.

The honest risk. The cheap first look might discard a tile that had a person in it. That failure is silent from the air and cannot be undone later. This is exactly why we intend to test it before adopting it, and why our pass mark assumes the pessimistic case: that such mistakes repeat from frame to frame rather than cancelling out.

4. Deliverables

1. **Three working aircraft**, flying as a coordinated group under one operator with no network dependency.
2. **A ground station** showing all three aircraft, their health, search progress and confirmed survivor positions — and deliberately *unable* to fly the aircraft, so the autonomy cannot be quietly undermined by an operator.
3. **Autonomy software** for area division, path planning, task allocation and failsafe behaviour, with an automated test suite.
4. **A measured detection figure** to replace the calculated one we rely on today. It would be the first real number we have.
5. **An evidence-based answer on flight altitude**, which our own design currently leaves unsettled.
6. **A possible publication.** Three research areas — image tiling, aerial person detection, and low-power onboard processing — have not been combined for flood conditions. That is a narrow, defensible contribution, and it is testable rather than merely argued.
7. **Code, data and results published either way.** A negative result still settles an open question and costs the same to obtain.

5. Benefits

Operationally, the system turns a slow, dangerous, manual search into a fast and systematic one. Three aircraft sweep ten hectares in under eight minutes and hand a rescue coordinator a list of coordinates rather than a video feed somebody has to watch. Crucially it does this *without* a network, which is the condition that actually obtains after a flood.

Economically, the whole three-aircraft system costs less than a single hour of helicopter search time. It travels by road, stores between seasons, and is operated by two people.

For the department, the programme leaves behind more than three aircraft: a calibrated thrust stand, a certified scale, a working ground station, an autonomy codebase with an automated test suite, and a group of students who have taken a system from arithmetic to flight. Most of that equipment, and all of the software, outlives this project.

Strategically, the design favours Indian suppliers wherever they can meet the specification — for sovereignty, and for the practical reason that domestic lead times are roughly half those of imports, which directly buys flight-test days inside a fixed schedule.

Beyond floods, nothing in the system is flood-specific except the training data. The same aircraft, planner and delivery mechanism apply to earthquake rubble survey, wildfire perimeter mapping, avalanche search and routine infrastructure inspection.

6. Resources needed

Money is not needed in one decision. The programme splits into **nine small steps**, and the order is engineering rather than accounting: prove the motor before buying twelve of them, and fly one complete aircraft before building the other two.

Each step is released only when the previous one has produced a result, so the department is never committed to more than one step at a time. **The first decision is INR 31,371** — about four per cent of the programme — and no single step exceeds INR 1,05,172.

All prices are current Indian retail. Totals include customs duty, GST and 15 % contingency where they apply.

#	Step	INR	Released only when
1	Prove the propulsion	31,371	On approval
2	Aircraft one: avionics and sensing	1,05,172	**Measured thrust $\geq$ 3.18 kgf per motor**
3	Aircraft one: airframe and power	89,192	Avionics powered and communicating on the bench
4	Fly aircraft one	42,574	Airframe assembled and weighed
5	Ground station and registration	66,992	First flight completed
6	Aircraft two: avionics	1,05,172	Aircraft one has flown a full mission
7	Aircraft two: airframe	94,367	Aircraft two's avionics verified
8	Aircraft three: avionics	1,05,172	Two aircraft flying, separation verified
9	Aircraft three: airframe	94,367	Aircraft three's avionics verified
	Total	**7,34,379**	

Step 1 — Prove the propulsion INR 31,371

The smallest step, and the one that decides everything after it. Our motors are sold without a published thrust curve, so we do not yet know that the aircraft can lift itself. Rather than buy twelve on faith, we buy *one* and measure it.

Item	Model	Qty	Total	Why
Calibrated scale	Bench scale, 30 kg × 1 g, certified	1	12,000	Mass is a hard limit; we cannot argue with a scale we do not own
Thrust stand	20 kg load cell + HX711 + power meter, frame built in-house	1	8,000	The instrument that answers the question
Motor	Tarot TL96020, 5008 / 340 KV	**1**	4,500	One, not twelve. Must lift 3.18 kgf

If this step fails, the programme stops here having spent INR 31,371 — and both instruments remain useful departmental equipment.

Step 2 — Aircraft one: avionics and sensing   INR 1,05,172

Item	Model	Qty	Total	Why
Flight controller	Holybro Pixhawk 6C Mini	1	22,600	Dual sensors, industry standard, open-source autopilot
GNSS receiver	Centimetre-accurate (RTK)	1	18,000	Position accuracy is the largest single error in the system
Camera + lens	Arducam IMX477 + 6 mm, **fixed focus**	1	11,100	Fixed focus is correct at every altitude we fly
AI accelerator	Raspberry Pi AI HAT+ (Hailo-8)	1	11,000	Runs the detector on board
Radio + storage	ExpressLRS 2.4 GHz	1	8,500	One link carries control *and* telemetry
Onboard computer	Raspberry Pi 5, 8 GB	1	8,000	The host everything runs on
Mesh radio	5.8 GHz + antennas	1	6,000	Video and data between aircraft

Step 3 — Aircraft one: airframe and power   INR 89,192

Item	Model	Qty	Total	Why
Motors	Tarot TL96020, remaining three	3	13,500	Bought only after step 1 proved the first
Airframe	25 × 23 mm carbon tube (Kineco Kaman) + machined clamps	1	13,000	Fabricated in the institute machine shop
Battery cells	Molicel P45B 21700, 4500 mAh	18	12,600	Peak draw is 38.3 A per cell; a 40 A cell has no margin
Pack assembly	6S 60 A BMS + power board + regulator	1	8,500	18 cells must be welded, fused and balanced
Speed controllers	50–60 A continuous, 6S	4	7,200	Peak is 29 A per motor
Payload release	4 metal-gear servos + mechanical detent	1	4,500	A power glitch must not drop a kit
Propellers	Tarot 1855 carbon, CW/CCW	4	4,000	See note below
Wiring	10 AWG silicone, XT90-S, JST-GH	1	2,400	Gauge is set by the 115 A peak
Propeller adapters	6 mm self-tightening hubs	4	1,400	A nut loosening at 5,000 rpm loses the aircraft

Step 4 — Fly aircraft one INR 42,574

Item	Model	Qty	Total	Why
Battery charger	**ToolkitRC M6D**, 500 W / 25 A, dual	2	11,672	See note below
Consumables	Fasteners, thread-lock, tape, filament	—	13,000	Consumed continuously; running out stops work
Safety-pilot radio	**RadioMaster Boxer**, ExpressLRS	1	6,955	A pilot taking over manually needs full-size sticks
Sun hood	Fabricated, plus a second-hand monitor	1	1,500	Screens are unreadable in Punjab daylight

Step 5 — Ground station and registration INR 66,992

Brought forward ahead of the second and third aircraft, because centimetre positioning has to be working before we can measure how accurately aircraft one reports a person's coordinates.

Item	Model	Qty	Total	Why
Field consumables	Cables, mounts, field spares	—	25,000	Corrections ride the existing radio, so no extra base radio
Base station	Second GNSS receiver on a tripod	1	18,000	Without it the aircraft have no corrections
Registration	Statutory, unmanned aircraft	3	5,000	At 6.36 kg these are Small category under the Drone Rules 2021
Survey tripod	Photographic tripod and adapter	1	4,000	Survey-grade is not required here

Steps 6 to 9 — Aircraft two and three

Each remaining aircraft repeats steps 2 and 3: first its avionics (INR 1,05,172), then its airframe and power (INR 94,367, slightly more than aircraft one because all four motors are bought together). The parts and the reasons are identical to the tables above.

These four steps are deliberately last. By the time any of them is requested, a complete aircraft has already flown the mission — so what is being funded is **replication of a design that works**, not a hope that it will.

Two notes on the parts

A saving worth having. The usual charger is the HOTA D6 Pro at INR 10,499–INR 13,500, but at 325 W per channel it can barely charge our packs. The ToolkitRC M6D is 500 W at INR 5,836 — cheaper and faster. It needs a separate power supply, which the department already owns. Two units give four channels, so the fleet charges at once on a flight day, and it still comes in INR 2,328 under budget.

One line is under-budgeted, and we would rather say so now. We carry INR 1,000 per propeller. The verified Indian listing is INR 3,250 per counter-rotating *pair*, so four propellers is INR 6,500 per aircraft rather than INR 4,000 — about INR 7,500 short across three aircraft, on the item most often broken in testing. We will look for a cheaper supplier first and absorb the rest from the charger saving and contingency; if neither works, this line has to rise.

What splitting this way costs

Smaller steps are not free. Buying one aircraft at a time means placing the same imported order three times, and imported parts take four to six weeks. Ordering all three sets together would save roughly **eight to twelve weeks** of schedule, at the price of a much larger single decision.

We have chosen the smaller steps because we would rather move slowly with your confidence than quickly without it. If schedule becomes the binding constraint, consolidating steps 2, 6 and 8 into one order is the lever — and that is your call, not ours.

For the detection test specifically, we need nothing beyond what the department already has: about **one week on a GPU** and **30 GB of storage**. The dataset is public and the software is open source.

7. Future work

Near term

Measure what is currently modelled. Detection accuracy, motor thrust and setup time are all calculated rather than observed. Each can be measured with equipment already in the plan above, and the programme is arranged so that every measurement arrives before the decision that depends on it.

Better sensing for water. People in water are easier to see in near-infrared than in ordinary colour: water absorbs infrared strongly and a person does not, so they appear bright against a dark background. A camera without its infrared filter costs about INR 3,000, and a public dataset exists to test the idea before buying anything. This is the cheapest promising experiment we have identified.

Beyond floods. Nothing in the aircraft or the planner is flood-specific. Earthquake rubble survey, wildfire perimeter mapping and avalanche search need the same capability with different training data.

Longer term: aircraft that must think for themselves

There is a class of problem where our central constraint — *a capable aircraft deciding for itself inside a very small power budget* — is not a design preference but a hard requirement. Planetary aviation is the clearest example.

NASA's *Ingenuity* flew 72 times on Mars before rotor damage grounded it in January 2024, from a 1.8 kg airframe with 1.21 m rotors, in an atmosphere less than one per cent as dense as Earth's. NASA's *Dragonfly*, launching in 2028 for Saturn's moon Titan, is a very different machine: about 875 kg on eight rotors, flying in an atmosphere four times denser than Earth's at one seventh the gravity — conditions far kinder to a rotorcraft than Mars.

What makes those vehicles hard is not the flying. It is that **a radio command takes minutes to arrive**, so nobody is piloting and nothing can be decided on the ground in time. Every choice — where to look, what is worth photographing, when to abort — has to be made on board, by a computer running on a power budget far smaller than ours.

That is our problem with the difficulty turned up, and the parts of this work that would transfer are the parts that are not about floods:

- deciding a survey pattern on board and executing it with no operator;
- running perception inside a fixed *energy* budget rather than to a fixed accuracy target;
- failing safely when the link is gone — for us a contingency, for them the normal condition.

We should be careful not to overstate this. Our own analysis found that the very smallest class of processor — the microcontroller-scale hardware often proposed for such missions — cannot see our targets at all, because shrinking the image to fit that processor destroys the thing we are looking for. A planetary version would need a different operating point: lower altitude, a larger target, or a coarser task. That is a genuine research question rather than a slide, and it is the honest version of the claim.

8. Conclusion

The design is further along than it looks. The sizing, the error budgets and the flight software are written down, reproducible, and explicit about which numbers are measured and which are still calculated.

What we are asking for is the means to finish it, released in nine steps so that nothing is committed before the previous step has produced evidence. The first step is INR 31,371 and answers the question everything else depends on: can these motors lift this aircraft.

The immediate technical need is smaller still — about a week of GPU time to test the detection idea on free public data. We would rather find out now whether it holds than build on a figure we calculated but never measured. Having already found one sizing mistake that way, testing first seems clearly the better choice.

We would also value your guidance on four decisions that are ours to make but would rather not make alone: the body size we design against, the operating altitude, whether to adopt the two-stage detection approach, and which of two processing modules to buy.